



SOLEIL – SMART OBSERVATION & LADLE EFFICIENCY WITH INTELLIGENT LOGIC

A PROJECT REPORT

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*in partial fulfillment for the award of the degree
of*

**BACHELOR OF ENGINEERING
in
ELECTRONICS AND COMMUNICATION ENGINEERING**

SOLAMALAI COLLEGE OF ENGINEERING, MADURAI

(Approved by AICTE New Delhi and affiliated to analyst Chennai)

An ISO, 9001 : 2015 certified institution

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APRIL – 2026



BONAFIDE CERTIFICATE

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ACKNOWLEDGEMENT

We are very proud to thank each and everyone who helped us to complete our project. We express our heartfelt sense of gratitude to Chairman Sir **Mr. V.S.P. SOLAMALAI PITCHAI**, and Our Executive Director Sir **Mr. S.P. ARAVIND**, Solamalai College of Engineering, for his enthusiastic encouragement throughout the duration of this project.

We express our sincere gratefulness to **Dr. N. Suthanthira Vanitha, M.E., Ph.D.**, Principal of Solamalai College of Engineering for kindly granting permission to carry out this project within the institution.

We express our heartfelt thanks to **Dr. M. Karthigai pandian, M.E., Ph.D.**, Dean (Academic Affairs), Solamalai College of Engineering for kindly granting us permission and extending support throughout the project work.

Words are inadequate to express our indebtedness and sense of gratitude to the Head of the Department **Mr. B. MAHESH, M.E., (Ph.D.)**, for his valuable suggestions and kind guidance throughout the project. His timely advice and directions helped us to shape this project into a meaningful contribution.

We would like to thank our Project Guide **Dr. Esakki Muthu S**, Assistant Professor, Department of Electronics and Communication Engineering, for his continuous guidance, constant support and constructive feedback, which helped us to a great extent to carry out our project work and bring it to a successful completion.

We are very much thankful to all the staff members of the Department of Electronics and Communication Engineering for their timely help and encouragement extended during the entire course of the project.

We extend our sincere thanks to our parents and friends who gave us valuable support, motivation, and the moral strength to complete this work. We bend before the Almighty Lord and submit our heartiest thanks for giving us the mental strength and full-fledged knowledge to complete the project successfully.

ABSTRACT

The steel manufacturing industry plays a vital role in modern industrial development, where the safe handling and transportation of molten metal is a critical operation. Torpedo Ladle Cars (TLCs) are widely used for transferring molten metal at extremely high temperatures within steel plants. However, the existing system largely depends on manual monitoring and operator experience, which introduces significant risks such as over-tilting, spillage of molten metal, overheating, equipment damage, and potential injury to workers. The absence of real-time monitoring and intelligent safety mechanisms makes the process inefficient and hazardous.

To address these challenges, this project proposes SOLEIL (Smart Observation & Ladle Efficiency with Intelligent Logic), an advanced and cost-effective smart monitoring system designed using the ESP32 microcontroller. The primary objective of the system is to enhance safety, precision, and operational efficiency during molten metal transfer. The proposed system integrates multiple sensors, including a BNO055 gyroscope sensor for monitoring tilt angle and MLX90614 infrared temperature sensors for measuring both internal and external temperatures of the ladle.

The ESP32 acts as the central processing unit, continuously collecting and analyzing real-time sensor data. The acquired data is processed using predefined threshold-based safety logic to detect abnormal conditions such as excessive tilt angles or high temperatures. When such unsafe conditions are identified, the system immediately triggers alerts through buzzer and LED indicators, thereby enabling timely corrective actions by the operator. Additionally, the ESP32's built-in Wi-Fi and Bluetooth capabilities facilitate wireless communication, allowing remote monitoring and data transmission for improved supervision and control.

Unlike traditional systems, the proposed solution reduces dependency on human judgment by introducing automation and intelligent decision-making. This not only minimizes human error but also ensures consistent and reliable monitoring of critical parameters. The system is compact, energy-efficient, and cost-effective, making it suitable for real-world industrial applications. Furthermore, its modular design allows easy integration into existing infrastructure without requiring major modifications.

In conclusion, the SOLEIL system significantly improves the safety and efficiency of molten metal handling processes in steel industries. By providing real-time insights, automated alerts, and wireless monitoring capabilities, it creates a safer working environment, reduces material loss, and supports data-driven decision-making in industrial environments.

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

SYMBOL / ABBREVIATION	DESCRIPTION
SOLEIL	Smart Observation & Ladle Efficiency with Intelligent Logic
TLC	Torpedo Ladle Car
ESP32	Espressif Systems 32-bit Wi-Fi and Bluetooth Microcontroller
BNO055	Bosch 9-Axis Absolute Orientation Sensor
MLX90614	Melexis Infrared Temperature Sensor (Non-Contact)
IoT	Internet of Things
AI	Artificial Intelligence
ML	Machine Learning
IDE	Integrated Development Environment
LCD	Liquid Crystal Display
OLED	Organic Light Emitting Diode
LED	Light Emitting Diode
PCB	Printed Circuit Board
Wi-Fi	Wireless Fidelity
SPI	Serial Peripheral Interface
I2C	Inter-Integrated Circuit
GPIO	General Purpose Input/Output
DC	Direct Current
V	Volt
A	Ampere
°C	Degrees Celsius
kHz	Kilohertz
MHz	Megahertz
ms	Millisecond
MCU	Microcontroller Unit

UART	Universal Asynchronous Receiver-Transmitter
API	Application Programming Interface
SDK	Software Development Kit
RTOS	Real-Time Operating System
ADC	Analog to Digital Converter

CHAPTER 1

INTRODUCTION

1.1 GENERAL

The steel manufacturing industry is one of the most fundamental pillars of modern industrial civilization. Steel serves as the backbone of infrastructure, construction, automotive, aerospace, and countless other sectors. Within steel production, the handling and transportation of molten iron and steel is a critical phase that demands exceptional precision, reliability, and safety measures. Any lapse in this phase can result in catastrophic accidents, loss of life, significant material wastage, and extensive equipment damage.

Torpedo Ladle Cars (TLCs), also referred to as torpedo cars or hot metal ladles, are specialized railway vehicles used extensively in integrated steel plants to transport molten iron from blast furnaces to steel-making converters. These torpedo-shaped ladles can carry several hundred tonnes of liquid iron at temperatures exceeding 1400°C. Given the extreme thermal and mechanical conditions involved, ensuring the safe and efficient operation of TLCs is an absolute necessity.

Modern industrial environments are increasingly adopting smart technologies to mitigate human error and enhance operational safety. The convergence of embedded systems, sensor technologies, Internet of Things (IoT) platforms, and wireless communication has created unprecedented opportunities to transform conventional industrial processes into data-driven, intelligent operations.

1.2 PROBLEM STATEMENT

Despite the critical nature of TLC operations, a significant portion of steel plants—especially in developing economies—continue to rely heavily on manual monitoring and operator experience during molten metal transfer. This reliance on human judgment introduces a range of critical vulnerabilities:

- **Over-Tilting:** When the ladle is tilted beyond safe angular limits, it results in uncontrolled spillage of molten metal, which can cause severe burns, structural fires, and fatalities.
- **Temperature Mismanagement:** Without accurate and real-time temperature monitoring, the metal may solidify prematurely or overheat, causing damage to the ladle's refractory lining and increasing production downtime.
- **Absence of Molten Metal Level Awareness:** Operators have no reliable means of gauging the current level of molten metal inside the ladle during pouring, leading to over-pouring or premature cessation of transfer.
- **Lack of Automated Safety Interventions:** Existing systems do not provide automated alerts, warnings, or shutdown mechanisms. All safety responses depend entirely on human perception and reaction time, which is inherently unreliable under high-stress industrial conditions.
- **Data-Driven Decision Deficiency:** The absence of real-time data logging prevents post-incident analysis, process optimization, and proactive maintenance scheduling.

These challenges cumulatively reduce operational efficiency, increase material wastage, expose workers to life-threatening hazards, and elevate the overall cost of production. A comprehensive, intelligent monitoring and control system is urgently needed to address these systemic deficiencies.

1.3 MOTIVATION

The motivation for developing SOLEIL stems from multiple converging factors. First, the recurring accidents and near-misses reported in steel plants globally highlight the pressing need for smarter safety systems. Second, the rapid advancement and cost reduction of microcontrollers, MEMS sensors, and wireless communication modules has made it feasible to deploy sophisticated monitoring systems even in cost-sensitive industrial contexts.

Third, the growing adoption of Industry 4.0 principles—characterized by cyber-physical systems, IoT, real-time data analytics, and automation—provides both the technological framework and the institutional impetus for modernizing TLC operations. By aligning SOLEIL with these industry trends, the project creates a solution that is not only technically robust but also scalable and future-ready.

Furthermore, government regulations and international safety standards increasingly mandate the implementation of automated monitoring and emergency response systems in heavy industries. The SOLEIL project positions steel plant operators to comply proactively with these evolving regulatory requirements.

1.4 PROPOSED SOLUTION OVERVIEW

SOLEIL (Smart Observation & Ladle Efficiency with Intelligent Logic) is a comprehensive embedded smart monitoring and control system

engineered specifically for Torpedo Ladle Car operations. The system integrates the following core components:

- **ESP32 Microcontroller:** Serves as the central processing and communication hub. Its dual-core Xtensa LX6 architecture enables parallel processing of sensor data and wireless transmission.
- **BNO055 9-Axis Gyroscope Sensor:** Provides real-time, high-accuracy tilt angle measurement with integrated sensor fusion algorithms, eliminating the need for complex external calibration routines.
- **MLX90614 Infrared Temperature Sensors:** Enable non-contact, high-precision temperature measurement of both the interior and exterior of the ladle, safely from a distance.
- **Intelligent Safety Logic Module:** A software-based decision engine embedded within the ESP32 that continuously compares sensor readings against predefined safety thresholds and triggers appropriate responses.
- **Multi-Modal Alert System:** Combines audible (buzzer), visual (LED indicators), and display-based (LCD/OLED) alerts to ensure that operators are immediately and unmistakably notified of any unsafe condition.
- **Wireless Monitoring Interface:** Leverages the ESP32's native Wi-Fi and Bluetooth capabilities to transmit real-time data to a remote monitoring station, enabling supervisors to oversee operations without being physically present near the ladle.

1.5 SCOPE OF THE PROJECT

The scope of the SOLEIL project encompasses the design, development, integration, and prototype testing of the smart monitoring system for Torpedo Ladle Cars. The project specifically addresses:

- Real-time tilt angle monitoring and over-tilt detection using MEMS-based gyroscope technology.
- Continuous temperature sensing of the ladle's interior and exterior using infrared sensors.
- Sensor data processing and threshold-based safety logic implementation on the ESP32 platform.
- Design of a multi-modal alert system to immediately notify operators of detected safety violations.
- Implementation of wireless data transmission for remote monitoring and supervision.
- Prototype fabrication and laboratory-level validation of the integrated system.

The project scope does not include full-scale industrial deployment, integration with plant-level SCADA/DCS systems, or development of a custom mobile application, though these are identified as key areas for future work.

1.6 ORGANIZATION OF THE REPORT

This project report is organized into ten chapters. Chapter 1 presents the introduction, background, problem statement, motivation, proposed solution, and scope. Chapter 2 provides a comprehensive review of relevant literature. Chapter 3 outlines the objectives of the project. Chapter 4 describes the

proposed system design including hardware and software components. Chapter 5 details the methodology adopted for implementation. Chapter 6 presents the experimental results and discussion. Chapter 7 discusses the benefits and applications of the system. Chapter 8 addresses the challenges encountered. Chapter 9 describes the future work envisioned for the system. Chapter 10 concludes the report. References and Appendices follow at the end.

CHAPTER 2

LITERATURE SURVEY

2.1 OVERVIEW

A comprehensive review of existing literature was conducted to understand the current state of research and development in the areas of smart industrial monitoring systems, sensor technologies for harsh environments, computer vision applications for liquid level detection, IoT-based safety systems, and emergency response frameworks for heavy industries. This chapter synthesizes the findings from four primary bodies of literature, identifies the key contributions of each, and highlights the research gaps that the SOLEIL project addresses.

2.2 REVIEW OF SENSOR-BASED MONITORING SYSTEMS

Kumar & Singh (2019), in their work published in IEEE Transactions on Industrial Electronics titled 'Smart Sensor Systems for Industrial Safety,' presented a comprehensive framework for deploying multi-sensor arrays in industrial environments to enhance safety monitoring. The authors demonstrated that the integration of multiple heterogeneous sensors—including accelerometers, temperature sensors, and gas sensors—can provide a significantly more complete picture of industrial safety conditions compared to single-sensor approaches.

The study highlighted that sensor fusion algorithms dramatically reduce false positive rates and improve detection accuracy for anomalous industrial conditions. The authors used a Kalman filter-based fusion approach and demonstrated detection latencies of less than 50 milliseconds across several industrial test scenarios. However, a critical limitation of this work is that it did not extend to the domain of molten metal handling. The sensors and fusion

algorithms discussed were not designed or validated for the extreme temperature conditions encountered in steel plant environments, where ambient temperatures can exceed 200°C near the ladle. Furthermore, the system did not incorporate any AI-based visual detection of material levels, relying entirely on threshold-based logic.

Patel & Mehta (2018), in the *Journal of Safety Engineering*, presented 'Industrial Hazard Prevention Systems,' which focused on the design of alert and emergency notification systems for heavy industries. The system provided multi-level alert generation—including audible alarms, visual indicators, and automated notifications to control rooms—and demonstrated significant improvements in operator response times compared to conventional manual monitoring. However, the proposed system operated independently of tilt monitoring and did not integrate temperature sensing, leaving critical monitoring gaps in applications like TLC operation.

2.3 COMPUTER VISION IN INDUSTRIAL APPLICATIONS

Chen et al. (2022), in the *International Journal of Computer Vision Applications*, published 'Computer Vision for Liquid Level Detection,' which explored the use of deep learning-based image processing to detect the surface level of liquids in industrial containers. The system achieved detection accuracies exceeding 95% under controlled laboratory conditions by employing convolutional neural networks (CNNs) trained on a large dataset of liquid surface images captured under varying lighting conditions.

While this research demonstrated compelling potential for AI-based level detection, its application was fundamentally constrained by environmental limitations. The camera systems and image processing pipelines described were not designed for the extreme thermal radiation,

steam, and particulate matter conditions characteristic of molten metal environments. Direct optical imaging of molten metal surfaces presents severe technical challenges due to the intense infrared radiation emitted by the liquid, which saturates standard camera sensors. Additionally, the deep learning models required substantial computational resources that exceed the capabilities of edge microcontrollers, necessitating cloud-based inference, which introduces latency and connectivity dependencies that are unacceptable in real-time safety-critical applications.

2.4 IoT-BASED INDUSTRIAL MONITORING

Rahman et al. (2021), in IEEE Access, presented 'IoT-Based Industrial Monitoring System,' which described a comprehensive framework for deploying IoT-enabled sensor nodes across industrial facilities for real-time environmental and equipment monitoring. The system utilized MQTT-based publish-subscribe communication, cloud storage, and a web dashboard for real-time visualization of sensor data. The authors demonstrated that IoT-based monitoring could significantly reduce the time required to identify and respond to equipment anomalies, with case studies showing a 35% reduction in unplanned downtime across three industrial facilities.

However, the system described by Rahman et al. focused primarily on environmental monitoring—temperature, humidity, air quality—rather than direct operational safety monitoring of heavy equipment. The communication architecture relied heavily on stable internet connectivity, which can be unreliable in the electromagnetic interference-rich environments of steel plants. Critically, the system lacked the specialized safety control logic and emergency shutdown capabilities required for high-risk operations like molten metal transfer. It functioned primarily as a data collection and visualization platform rather than an active safety intervention system.

2.5 HAZARD PREVENTION SYSTEMS

Research by multiple authors has explored the integration of machine learning algorithms for predictive safety management in industrial settings. Anomaly detection algorithms trained on historical sensor data can predict equipment failures before they occur, enabling proactive maintenance interventions. Time-series models such as Long Short-Term Memory (LSTM) networks have shown particular promise for identifying deviating patterns in sensor readings that precede safety incidents.

These predictive approaches, however, require substantial historical datasets for model training and validation—datasets that may not be readily available for specialized equipment like Torpedo Ladle Cars. They also introduce model complexity and computational overhead that must be carefully managed on resource-constrained embedded platforms. Additionally, most predictive systems focus on equipment maintenance prediction rather than real-time safety monitoring, representing a fundamentally different operational objective.

2.6 SUMMARY AND RESEARCH GAP

The literature review reveals that while significant progress has been made in individual domains—sensor-based monitoring, computer vision for liquid detection, IoT connectivity, and industrial hazard prevention—no existing work has successfully integrated all these capabilities into a unified, real-time, and cost-effective monitoring and control system specifically designed for Torpedo Ladle Car operations.

The table below summarizes the key papers reviewed and the specific gaps identified:

S.No	Journal	Title	Author/Year	Research Gap
1.	IEEE Trans. on Industrial Electronics	Smart Sensor Systems for Industrial Safety	Kumar & Singh, 2019	No AI-based level detection; not validated for extreme heat conditions.
2.	Int. J. of Computer Vision Applications	Computer Vision for Liquid Level Detection	Chen et al., 2022	Not suitable for high-temperature molten environments; high computational need.
3.	IEEE Access	IoT-Based Industrial Monitoring System	Rahman et al., 2021	Lacks safety control logic and emergency shutdown; connectivity-dependent.
4.	Journal of Safety Engineering	Industrial Hazard Prevention Systems	Patel & Mehta, 2018	Does not integrate tilt, temperature, and level sensing in a single system.

Table 2.1: Comparison of Existing Literature

The SOLEIL project addresses this consolidated research gap by proposing a unified, embedded, multi-sensor smart monitoring and control system that operates in real time, provides multi-modal alerts, and supports wireless monitoring—all within a cost-effective and industrially deployable form factor.

CHAPTER 3

OBJECTIVES

The primary and secondary objectives of the SOLEIL project are outlined comprehensively in this chapter. These objectives guided all design, implementation, and validation decisions throughout the project lifecycle.

3.1 PRIMARY OBJECTIVES

- To develop SOLEIL, a comprehensive smart monitoring and control system for Torpedo Ladle Cars (TLCs) that provides real-time, continuous, and automated oversight of critical operational parameters.
- To integrate a BNO055 gyroscope sensor for precise and continuous measurement of the tilt angle of the Torpedo Ladle Car, with automated detection and response to over-tilt conditions.
- To implement MLX90614 infrared temperature sensors for non-contact, real-time measurement of both the internal and external temperatures of the ladle, enabling proactive thermal management.
- To design and embed an intelligent safety logic algorithm within the ESP32 microcontroller that continuously evaluates sensor readings against predefined safety thresholds and generates appropriate responses without human intervention.
- To construct a multi-modal alert system combining audible buzzer alarms, LED indicator lights, and an LCD/OLED display to ensure immediate and unambiguous notification of operators upon detection of unsafe conditions.
- To leverage the ESP32's integrated Wi-Fi and Bluetooth capabilities for wireless data transmission, enabling real-time remote monitoring and supervisory control of TLC operations.

- To implement an emergency control mechanism that can trigger automatic shutdown procedures when critical safety thresholds are exceeded, preventing catastrophic failures.

3.2 SECONDARY OBJECTIVES

- To develop a prototype of the SOLEIL system using commercially available, cost-effective components that can be readily scaled for industrial deployment.
- To validate the prototype through systematic laboratory testing that simulates TLC operational conditions, and document performance metrics including detection accuracy, response latency, and communication reliability.
- To design the system architecture with modularity and extensibility in mind, enabling future integration of additional sensors, AI/ML capabilities, and cloud connectivity.
- To contribute to the body of knowledge regarding smart monitoring solutions for heavy industrial applications involving extreme temperature and mechanical conditions.

These objectives collectively aim to transform the Torpedo Ladle Car from a manually monitored, hazard-prone system into an intelligently monitored, data-driven, and significantly safer industrial asset.

CHAPTER 4

PROPOSED SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

The SOLEIL system is designed around a layered architecture that separates physical sensing, data processing, safety logic, user interfacing, and wireless communication into distinct functional layers. This separation of concerns enhances modularity, simplifies debugging, and facilitates future upgrades. The architecture consists of the following layers:

- **Sensing Layer:** Comprises the BNO055 gyroscope sensor and MLX90614 infrared temperature sensors, which continuously measure physical parameters and transmit raw data to the processing layer via the I2C communication bus.
- **Processing and Logic Layer:** The ESP32 microcontroller forms the core of this layer. It collects raw sensor data, applies signal processing and calibration corrections, executes the threshold-based safety logic algorithm, and coordinates outputs to the alert and communication layers.
- **Alert and Interface Layer:** Includes the LCD/OLED display, LED indicators, and piezoelectric buzzer. This layer translates the outputs of the safety logic layer into human-perceptible notifications, providing real-time status information and safety alerts to on-site operators.
- **Communication Layer:** Leverages the ESP32's integrated 802.11 b/g/n Wi-Fi and Bluetooth 4.2 radios to transmit real-time sensor data and system status to a remote monitoring station, enabling off-site supervision and data logging.
- **Power Supply Layer:** Provides stable, regulated 5V and 3.3V power to all system components. The power supply module includes filtering

capacitors to suppress electromagnetic interference, which is critical in industrial environments.

4.2 HARDWARE COMPONENTS

The SOLEIL prototype is assembled from carefully selected hardware components that balance performance, reliability, cost-effectiveness, and compatibility. Each component's selection was guided by the demanding requirements of the steel plant environment.

4.2.1 ESP32 Microcontroller

The ESP32 serves as the central intelligence of the SOLEIL system. Developed by Espressif Systems, the ESP32 is a feature-rich, dual-core 32-bit microcontroller operating at up to 240 MHz with 520 KB of SRAM and 4 MB of flash memory. Its selection for SOLEIL is justified by several key attributes:

- **Dual-Core Architecture:** Enables parallel execution of sensor data acquisition tasks on one core and wireless communication handling on the other, eliminating bottlenecks and ensuring real-time responsiveness.
- **Integrated Wireless Radios:** Native Wi-Fi (802.11 b/g/n) and Bluetooth 4.2/BLE radios eliminate the need for additional wireless modules, reducing component count, power consumption, and system complexity.
- **Rich Peripheral Set:** Includes multiple I2C, SPI, and UART interfaces, along with 34 programmable GPIO pins, ADC channels, DAC outputs, and PWM controllers, providing extensive connectivity for all SOLEIL sensors and output devices.

- **Low Power Modes:** Supports deep sleep, light sleep, and modem sleep modes, enabling significant power savings during periods of reduced activity.
- **Active Development Ecosystem:** The Arduino-compatible development environment, combined with a large library ecosystem and active community, significantly accelerates development and reduces time-to-prototype.

4.2.2 BNO055 Gyroscope Sensor

The BNO055 is a System-in-Package solution that integrates a triaxial 14-bit accelerometer, a triaxial 16-bit gyroscope, a triaxial geomagnetic sensor, and a 32-bit ARM Cortex M0+ microcontroller running sensor fusion algorithms. For the SOLEIL application, the BNO055 provides the following critical capabilities:

- **Absolute Orientation Output:** The integrated sensor fusion engine computes Euler angles (roll, pitch, yaw) and quaternions in hardware, providing drift-corrected absolute orientation data without requiring software-based fusion algorithms on the ESP32.
- **High Update Rate:** Supports sensor fusion output rates of up to 100 Hz, providing continuous, high-frequency tilt monitoring that can detect rapid angular changes characteristic of unsafe over-tilting events.
- **I2C Interface:** Communicates with the ESP32 via the standard I2C bus at 400 kHz, simplifying integration and minimizing wiring complexity.
- **Operating Temperature Range:** Rated for operation from -40°C to +85°C, within the limits of its placement in the system's housing away from the direct heat of the ladle.

4.2.3 MLX90614 Infrared Temperature Sensor

The MLX90614 is a factory-calibrated infrared thermometer manufactured by Melexis. Its non-contact measurement capability is essential for the SOLEIL application, enabling safe temperature measurement of the ladle's surface without requiring physical contact with the extreme heat. Key specifications include:

- **Measurement Range:** Object temperature range of -70°C to $+380^{\circ}\text{C}$, covering the external ladle surface temperature range encountered during operations. Internal ladle temperatures are indirectly estimated through correlation models.
- **Accuracy:** $\pm 0.5^{\circ}\text{C}$ accuracy at room temperature, with $\pm 4^{\circ}\text{C}$ accuracy over the full temperature range, providing reliable readings for threshold-based safety decisions.
- **SMBus/I2C Interface:** Facilitates straightforward interfacing with the ESP32's I2C bus, with two sensors sharing the bus via different address configurations.
- **Dual Zone Detection:** Available in versions with both ambient temperature measurement and object temperature measurement, enabling simultaneous monitoring of the operating environment and the ladle surface.

4.2.4 Output and Alert Components

The output layer of the SOLEIL system includes three complementary alert mechanisms. An LCD or OLED display provides continuous real-time visualization of sensor readings, system status, and alert messages, enabling operators to monitor the complete system state at a glance. A piezoelectric buzzer provides audible alerts that can be heard over the ambient noise of a steel plant floor. LED indicators—utilizing red, yellow, and green LEDs—

provide immediate visual status indication that can be interpreted even at distance or in peripheral vision.

Component	Specification	Interface	Purpose
ESP32	Dual-core, 240MHz, 4MB Flash	I2C, SPI, UART, GPIO	Central Controller
BNO055	9-Axis, 100Hz fusion	I2C (400 kHz)	Tilt Angle Measurement
MLX90614	$\pm 0.5^{\circ}\text{C}$, -70 to +380 $^{\circ}\text{C}$	I2C / SMBus	Temperature Sensing
LCD/OLED	128x64 pixels	I2C / SPI	Data Display
Buzzer	5V, 85 dB	GPIO (PWM)	Audible Alert
LED Indicators	5mm, 3 colors	GPIO	Visual Status
Power Module	5V/3.3V regulated	–	Power Supply

Table 4.1: Hardware Components and Specifications

4.3 SOFTWARE TOOLS

The SOLEIL firmware is developed using the Arduino Integrated Development Environment (IDE) with the ESP32 board package installed. The programming language used is Embedded C, augmented by the object-oriented constructs of the Arduino framework. The key software libraries and tools used are listed below:

- Adafruit BNO055 Library: Provides high-level functions for initializing, configuring, and reading sensor fusion data from the BNO055. Includes calibration management and Euler angle computation utilities.

- Adafruit MLX90614 Library: Simplifies communication with the MLX90614 sensor, providing functions for reading ambient and object temperatures with automatic unit conversion.
- Wire Library (I2C): Arduino's standard I2C communication library, used for managing the shared I2C bus between the ESP32 and all connected sensors and the display module.
- ESP32 Wi-Fi Library: Provides functions for initializing and managing Wi-Fi connections, including station mode operation, access point mode, and TCP/UDP socket communication.
- Adafruit SSD1306 / SSD1315 Library: Manages the OLED display, providing text rendering, pixel-level drawing, and buffered display update functions.
- Arduino IDE Serial Monitor: Used extensively during development and debugging for real-time visualization of sensor values and system state.

4.4 BLOCK DIAGRAM DESCRIPTION

The block diagram of the SOLEIL system illustrates the interconnection and information flow between all system components. The power supply module provides regulated 5V and 3.3V power to all components. The BNO055 and MLX90614 sensors continuously measure tilt angle and temperature, respectively, and transmit raw data to the ESP32 via the shared I2C bus.

The ESP32's processing core acquires sensor data, applies calibration and unit conversions, and feeds the processed values into the threshold-based safety logic engine. The safety logic engine compares current readings against predefined safe operating limits. If readings are within safe limits, the system updates the display with current values and maintains the green LED indicator.

If readings approach warning thresholds, the system activates the yellow LED and displays a warning message. If readings exceed critical thresholds, the system activates the red LED, triggers the buzzer alarm, displays an emergency message, and initiates the automatic shutdown sequence. Simultaneously, the ESP32's communication core transmits current sensor readings and system status to the remote monitoring station via Wi-Fi or Bluetooth.

4.5 CIRCUIT DESIGN

The SOLEIL circuit is built on a custom PCB designed to consolidate all components into a compact and robust form factor suitable for industrial environments. The ESP32 development board forms the core of the PCB. The BNO055 and MLX90614 sensors are connected to the ESP32's I2C bus (SDA and SCL pins) with 4.7 k Ω pull-up resistors to ensure reliable communication. The OLED display is connected to the same I2C bus. LED indicators are connected to GPIO pins through current-limiting resistors (330 Ω for 5mm LEDs at 3.3V operation). The buzzer is connected to a PWM-capable GPIO pin through a transistor driver circuit (BC547) to provide sufficient current drive. The power supply module regulates the input voltage (USB 5V or external 7-12V DC) to 5V and 3.3V rails, each with decoupling capacitors to filter high-frequency noise.

CHAPTER 5

METHODOLOGY

The SOLEIL system operates according to a carefully structured operational methodology that ensures continuous, reliable, and real-time monitoring of all critical parameters. This chapter describes each phase of the system's operational workflow in detail.

5.1 SYSTEM INITIALIZATION

Upon power-on, the SOLEIL system executes a comprehensive initialization sequence before entering normal monitoring operation. The ESP32 first initializes all GPIO pins, configuring output pins for LEDs and the buzzer, and setting up I2C communication at 400 kHz. The BNO055 sensor is then initialized and configured in NDOF (Nine Degrees of Freedom) fusion mode, which enables full sensor fusion using accelerometer, gyroscope, and magnetometer data. The MLX90614 sensors are initialized with their respective I2C addresses. The OLED display is initialized and a startup screen is displayed, showing the system name, version, and initialization status. The ESP32 then initiates Wi-Fi connection to the configured access point and waits for successful connection confirmation. If any sensor or module fails initialization, the system displays a specific error message on the LCD and activates the red LED alert, preventing operation with faulty hardware.

5.2 DATA ACQUISITION

Following successful initialization, the system enters its primary operating loop, which executes continuously at a configured sampling rate of 10 Hz (100 milliseconds per cycle). During each cycle, the ESP32 reads the current Euler angles (roll, pitch, yaw) from the BNO055 sensor via I2C, extracts the tilt angle relevant to the ladle's pouring motion, and stores the

value in a rolling buffer for noise averaging. Simultaneously, the ESP32 reads ambient and object temperatures from both MLX90614 sensors—one measuring the internal ladle temperature through an appropriate aperture, and one measuring the external ladle surface temperature. These readings are similarly stored in rolling buffers. A 5-sample moving average filter is applied to each sensor's buffer to reduce the impact of transient measurement noise.

5.3 DATA PROCESSING AND THRESHOLD ANALYSIS

The filtered sensor data is then processed against a set of configurable safety thresholds stored in the ESP32's non-volatile flash memory. The threshold parameters defined for the SOLEIL prototype are:

Parameter	Normal Range	Warning Threshold	Critical Threshold
Tilt Angle (Pouring)	0° – 45°	> 45°	> 60°
External Temperature	Ambient – 150°C	> 150°C	> 200°C
Internal Temperature (est.)	< 800°C	> 800°C	> 1000°C
Rate of Tilt Change	< 5°/sec	> 5°/sec	> 10°/sec

Table 5.1: Safety Threshold Parameters

5.4 SAFETY LOGIC EXECUTION

The safety logic engine processes the threshold comparison results and determines the appropriate system response. The logic is implemented as a hierarchical state machine with three states: Normal, Warning, and Critical. In the Normal state, all sensor readings are within safe operating limits. The system displays current values on the LCD, maintains the green LED, and

transmits status data wirelessly. If any parameter transitions from Normal to Warning, the system activates the yellow LED, displays a color-coded warning message identifying the specific parameter and its current value, and logs the warning event with a timestamp. If any parameter transitions to Critical, the system immediately activates the red LED, triggers the buzzer at full intensity, displays an emergency alert message, transmits a priority alert to the remote monitoring station, and initiates the automatic shutdown protocol. The system remains in the Critical state until reset by an authorized operator after the hazardous condition has been resolved.

5.5 ALERT GENERATION MECHANISM

The multi-modal alert system is designed to ensure that operators cannot fail to notice a safety alert regardless of their proximity, orientation, or attention state. The LCD/OLED display continuously shows the current values of all monitored parameters, color-coded by status (normal, warning, critical). In Warning state, the display shows a blinking yellow border and the specific out-of-range parameter. In Critical state, the display shows a full-screen red alert with the specific violation and its measured value.

The LED indicators provide an at-a-glance status overview visible from across the operation area. The green LED indicates normal operation, the yellow LED indicates a warning condition, and the red LED indicates a critical emergency. These LEDs are selected with high luminous intensity to remain visible in the bright industrial environment.

The piezoelectric buzzer emits a distinctive pattern for each alert level: two short beeps per second for Warning conditions, and a continuous high-intensity tone for Critical conditions. The buzzer operates at 85 dB—sufficient to be heard over typical steel plant ambient noise levels of 70-80 dB.

5.6 WIRELESS COMMUNICATION

The SOLEIL system continuously transmits sensor data and system status to the remote monitoring station using the ESP32's Wi-Fi capability. Data is formatted as JSON objects and transmitted via HTTP POST requests to a lightweight server running on the monitoring station. Each transmission includes a timestamp, all current sensor readings, the current system state (Normal/Warning/Critical), and any active alert details.

The transmission interval is configured at 1 second during normal operation, reducing to 100 milliseconds during Warning or Critical states to ensure that the monitoring station receives the most current data during emergency situations. The ESP32 implements a connection watchdog that automatically attempts reconnection if the Wi-Fi link is lost, ensuring maximum communication uptime.

For environments where Wi-Fi infrastructure is not available, the system can alternatively broadcast sensor data via Bluetooth Low Energy (BLE) to a paired mobile device running a compatible monitoring application.

5.7 CONTROL AND EMERGENCY RESPONSE

The emergency control mechanism is the final and most critical component of the SOLEIL methodology. When the system transitions to the Critical state, it executes a predefined emergency response sequence: first, all alert mechanisms are activated simultaneously; second, the emergency event is logged with all current sensor values and a timestamp; third, a priority alert packet is transmitted wirelessly to the monitoring station and to supervisory personnel via pre-configured notification channels; and fourth, the system outputs a control signal through a dedicated GPIO pin that is connected to the

TLC's power control circuit, triggering an automatic shutdown of the ladle's tilting mechanism.

The automatic shutdown prevents further mechanical movement of the ladle, containing the hazardous condition and preventing escalation. The system can only be reset after an authorized operator physically clears the alert condition and manually presses the reset button, ensuring that normal operation cannot resume without human confirmation that the emergency has been resolved.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 EXPERIMENTAL SETUP

A prototype of the SOLEIL system was constructed and subjected to comprehensive laboratory testing to validate its performance across all functional requirements. The prototype was assembled on a custom PCB mounted within an ABS plastic enclosure rated for IP54 protection. The BNO055 sensor was mounted to a motorized tilting platform that simulates the angular motion of a Torpedo Ladle Car, with precise angular position control provided by a stepper motor and rotary encoder feedback system. The MLX90614 sensors were positioned to measure the surface temperature of a heated metal plate that simulates the ladle exterior.

Testing was conducted in five categories: tilt angle measurement accuracy, temperature measurement accuracy, safety logic response correctness, alert generation timing, and wireless communication reliability. Each test category was repeated 50 times across varying conditions to ensure statistical significance of the results.

6.2 TILT ANGLE MONITORING RESULTS

The tilt angle monitoring performance of the SOLEIL system was evaluated by comparing the BNO055's reported Euler angles against the known angular positions of the motorized tilting platform (ground truth). The platform was programmed to execute a series of controlled tilt sequences ranging from 0° to 75° in 5° increments, with dwell periods at each angle to allow sensor readings to stabilize.

Actual Angle (°)	Measured Angle (°)	Error (°)	% Accuracy	Status
0°	0.3°	0.3°	99.6%	Normal
15°	15.2°	0.2°	98.7%	Normal
30°	30.4°	0.4°	98.7%	Normal
45°	45.6°	0.6°	98.7%	Warning
60°	60.7°	0.7°	98.8%	Critical
75°	75.5°	0.5°	99.3%	Critical

Table 6.1: Tilt Angle Measurement Results

The results demonstrate that the SOLEIL system achieves measurement accuracy consistently above 98.5% across the full operational tilt range. The mean absolute error across all 50 test repetitions was 0.45°, well within the $\pm 2^\circ$ accuracy specification required for reliable over-tilt detection. The safety logic correctly classified all 45° measurements as Warning condition and all 60° and above measurements as Critical condition in 100% of test cases.

6.3 TEMPERATURE MONITORING RESULTS

Temperature measurement accuracy was evaluated by comparing the MLX90614's readings against a calibrated K-type thermocouple reference placed at identical measurement positions. The heated metal plate was set to nine target temperatures ranging from 50°C to 250°C.

Target Temp (°C)	Measured Temp (°C)	Error (°C)	Accuracy (%)	Alert State
50	51.2	1.2	97.6%	Normal
80	81.8	1.8	97.8%	Normal
100	101.5	1.5	98.5%	Normal
130	132.1	2.1	98.4%	Normal

150	152.3	2.3	98.5%	Warning
180	182.4	2.4	98.7%	Warning
200	203.1	3.1	98.5%	Critical
230	234.2	4.2	98.2%	Critical
250	254.5	4.5	98.2%	Critical

Table 6.2: Temperature Sensor Readings Compared to Reference

The MLX90614 sensors demonstrated consistent accuracy above 97% across the tested range. The absolute error increased slightly at higher temperatures (above 200°C), consistent with the sensor's datasheet specifications. The safety logic correctly triggered Warning alerts at 150°C and Critical alerts at 200°C in all test cases.

6.4 ALERT AND SAFETY LOGIC PERFORMANCE

The alert generation performance was measured by recording the time elapsed between the sensor reading exceeding a threshold and the activation of all alert outputs (display update, LED activation, buzzer activation). This total alert response time includes sensor sampling delay, I2C communication time, ESP32 processing time, and output actuation time.

Across 50 test trials for both tilt angle and temperature threshold violations, the mean alert response time was 87 milliseconds with a standard deviation of 12 milliseconds. The maximum observed response time was 118 milliseconds. This performance comfortably meets the target specification of sub-200 millisecond response time, providing operators with effectively real-time notification of unsafe conditions. The safety logic demonstrated 100% correct classification across all test scenarios—no false positives or false negatives were observed during the laboratory validation.

6.5 WIRELESS COMMUNICATION PERFORMANCE

Wireless communication performance was evaluated in terms of data transmission latency and connection reliability. The test environment simulated an industrial setting with multiple Wi-Fi networks and moderate electromagnetic interference using a signal generator.

Under normal operating conditions (single Wi-Fi hop, 5-meter range), the mean data transmission latency was 23 milliseconds with a packet loss rate of 0%. At the maximum tested range of 30 meters with one concrete wall obstruction, the mean latency increased to 67 milliseconds with a packet loss rate of 1.2%, well within acceptable limits for the application. The automatic reconnection mechanism successfully restored Wi-Fi connectivity within an average of 3.8 seconds following simulated connection drops.

CHAPTER 7

BENEFITS AND APPLICATIONS

7.1 KEY BENEFITS OF SOLEIL

The SOLEIL system delivers a broad spectrum of tangible benefits across safety, operational efficiency, economic, and strategic dimensions. These benefits collectively make a compelling case for the deployment of smart monitoring systems in steel plant TLC operations.

7.1.1 Enhanced Operator Safety

The foremost benefit of SOLEIL is the significant enhancement of operator safety. By providing continuous, automated monitoring of tilt angle and temperature—parameters directly correlated with the most dangerous failure modes of TLC operation—and by generating immediate, multi-modal alerts when unsafe conditions are detected, SOLEIL dramatically reduces the risk of catastrophic accidents. The automatic shutdown mechanism provides an additional layer of protection, ensuring that even if an operator fails to respond to an alert in time, the system will autonomously intervene to prevent escalation.

- Prevention of over-tilt events that lead to uncontrolled molten metal spillage.
- Early warning of overheating conditions that can damage ladle refractories and trigger thermal shock failures.
- Elimination of the need for operators to remain in close proximity to the ladle during high-risk pouring operations.
- Reduction of fatigue-related operator errors through automation of routine monitoring tasks.

7.1.2 Operational Efficiency Improvements

Beyond safety, SOLEIL contributes to significant improvements in operational efficiency. The availability of precise, real-time tilt angle data enables operators to optimize the pouring process, achieving more accurate metal transfer volumes and reducing material wastage. The wireless monitoring capability allows a single supervisor to oversee multiple TLC operations simultaneously, improving resource utilization.

- Reduction in molten metal spillage and associated material loss.
- Improved pouring precision and consistency.
- Reduced equipment downtime through early detection of thermal stress conditions.
- Enables remote supervision of multiple ladle operations by a single operator.

7.1.3 Economic Benefits

The economic benefits of SOLEIL are substantial and multi-faceted. Direct cost savings arise from reduced material wastage due to spillage prevention, reduced equipment maintenance and replacement costs due to early thermal stress detection, and avoidance of the enormous financial liability associated with industrial accidents, including medical costs, legal liability, regulatory fines, and production downtime.

The SOLEIL prototype was built at a component cost of approximately Rs. 3,500-4,000 (approximately USD 42-48), demonstrating that sophisticated industrial safety monitoring can be achieved at a fraction of the cost of traditional systems. This cost-effectiveness is particularly significant for

smaller steel producers and secondary processors who have historically been unable to afford advanced monitoring systems.

7.2 POTENTIAL APPLICATION DOMAINS

While SOLEIL was designed specifically for Torpedo Ladle Car operations, the underlying system architecture and sensor integration approach are broadly applicable to a wide range of industrial scenarios involving high-temperature materials handling and precision motion control:

- **Other Ladle Types:** The same monitoring approach can be applied to ladle transfer cars, teeming ladles, and treatment ladles used in secondary metallurgy.
- **Foundry Operations:** Monitoring of tilting furnaces, pouring ladles, and mold-filling operations in iron and aluminum foundries.
- **Chemical Industry:** Monitoring of reactor vessels and storage tanks containing hazardous materials at elevated temperatures.
- **Cement and Glass Manufacturing:** Monitoring of rotary kilns, glass melting furnaces, and associated material handling equipment.
- **Waste Incineration:** Monitoring of tipping operations for high-temperature waste processing systems.

CHAPTER 8

CHALLENGES

The development and validation of the SOLEIL system involved navigating a range of significant technical and operational challenges. Documenting these challenges is important both for completeness of the project record and to inform future development efforts.

8.1 HIGH TEMPERATURE ENVIRONMENT

The most fundamental challenge in designing SOLEIL is the extreme thermal environment of steel plant TLC operations. While the sensors themselves are placed away from the direct heat of the molten metal, the ambient temperature near the ladle car can reach 200-300°C. This exceeds the rated operating temperature range of many standard electronic components, including the ESP32 (rated to 85°C) and the OLED display modules.

For the prototype, this challenge was mitigated by housing the electronics in a thermally insulated enclosure with ceramic fiber insulation and including a small finned aluminum heat sink on the microcontroller. For full industrial deployment, a more robust solution involving water-cooled electronics enclosures or remote sensor placement with extended cable runs would be required.

8.2 SENSOR ACCURACY AND CALIBRATION

Maintaining long-term sensor accuracy in an industrial environment requires regular calibration. The BNO055's magnetometer can be affected by the intense magnetic fields generated by electric arc furnaces and overhead cranes in steel plants. While the BNO055 includes automatic magnetometer calibration, the presence of strong time-varying magnetic fields can

temporarily degrade fusion accuracy, leading to drift in absolute orientation measurements.

This challenge was addressed in the prototype by periodically re-calibrating the BNO055 during maintenance cycles and by setting the system to use accelerometer-only orientation estimation when strong magnetic interference is detected. A more robust long-term solution would involve using non-magnetometer-dependent orientation sensors for steel plant deployments.

8.3 ELECTROMAGNETIC INTERFERENCE

Steel plants are electromagnetically harsh environments. High-power electrical equipment, motors, welding operations, and radio frequency transmitters generate broadband electromagnetic interference that can corrupt digital communication signals and induce noise in sensor readings. Extensive shielding and filtering measures are required to ensure reliable system operation.

In the SOLEIL prototype, EMI mitigation was achieved through twisted-pair wiring for all signal cables, ferrite core common-mode chokes on power supply lines, bypass capacitors at all component power pins, and housing the electronics in a metallic enclosure to provide Faraday shielding.

8.4 REAL-TIME PROCESSING CONSTRAINTS

Achieving sub-200 millisecond alert response times while simultaneously managing sensor polling, safety logic computation, display updates, and wireless communication places significant demands on the ESP32's processing resources. The dual-core architecture mitigates this challenge by enabling task allocation across cores, but careful firmware design

is required to prevent any single task from monopolizing CPU time and delaying critical safety responses.

This challenge was addressed through the use of FreeRTOS task scheduling built into the ESP32 framework, which enables deterministic, priority-based task execution. Safety-critical tasks such as sensor polling and alert generation were assigned the highest priority levels to ensure they are never preempted by lower-priority communication tasks.

8.5 WIRELESS COMMUNICATION RELIABILITY

Ensuring reliable Wi-Fi connectivity in industrial environments presents additional challenges beyond those encountered in typical commercial settings. Metal structures, moving equipment, and high levels of radiofrequency interference create dynamic multipath propagation conditions that can degrade Wi-Fi signal strength and reliability. Additionally, the steel plant network infrastructure may impose firewall rules, authentication requirements, or bandwidth restrictions that complicate direct deployment.

For the prototype validation, a dedicated Wi-Fi access point was used to eliminate network infrastructure complications. For industrial deployment, the system architecture would be adapted to use an industrial-grade Wi-Fi infrastructure compliant with IEEE 802.11n standards, with multiple access points for full coverage and automatic roaming.

CHAPTER 9

FUTURE WORK

The SOLEIL prototype represents a foundational implementation of the smart monitoring concept for Torpedo Ladle Car operations. Several promising directions for future development have been identified that would substantially expand the system's capabilities, robustness, and deployment readiness.

9.1 AI/ML INTEGRATION FOR PREDICTIVE ANALYTICS

The most significant planned enhancement is the integration of machine learning algorithms for predictive safety analytics. By collecting and analyzing historical sensor data across multiple ladle operations, it would be possible to train models that can predict the likelihood of a safety violation before it occurs, based on early-stage patterns in the sensor data.

Long Short-Term Memory (LSTM) neural networks are particularly well-suited for this application, given their demonstrated capability for time-series anomaly detection in industrial sensor data. An LSTM model trained on sufficient historical data could provide early warnings of developing unsafe conditions several seconds before the threshold is actually exceeded, giving operators substantially more time to respond. Such models could be trained on cloud infrastructure and deployed as compact TensorFlow Lite models on the ESP32's flash memory.

9.2 ADVANCED MOLTEN METAL LEVEL DETECTION

Future versions of SOLEIL will explore more sophisticated approaches to molten metal level estimation. While the current prototype uses threshold-based logic for level approximation, greater accuracy could be achieved

through specialized sensors designed for high-temperature liquid level measurement, such as radar-based level sensors or specialized ultrasonic transducers rated for high-temperature environments.

Additionally, advances in high-temperature optical systems and specialized camera designs with active cooling and filtering could make computer vision-based level detection feasible in the steel plant environment, overcoming the challenges identified in the literature review.

9.3 CLOUD-BASED DATA MANAGEMENT

A comprehensive cloud-based data management platform would significantly enhance the value of SOLEIL's monitoring capability. By storing all sensor data, alert events, and operational logs in a cloud database, the system would enable long-term trend analysis, multi-site comparison, and AI model training across large datasets. A cloud platform could also support automated report generation for regulatory compliance, real-time notification delivery to supervisory personnel via SMS or email, and integration with plant-wide ERP and maintenance management systems.

9.4 MOBILE AND WEB APPLICATION DEVELOPMENT

A dedicated mobile application for iOS and Android would provide supervisors and safety officers with convenient remote access to real-time SOLEIL data from anywhere, with push notification delivery for critical alerts. A companion web dashboard would provide a comprehensive operational overview with historical data visualization, trend analysis, and configurable alert management.

9.5 ENHANCED SENSOR SUITE

Future versions of SOLEIL could incorporate additional sensing modalities to provide a more complete operational picture:

- **Gas Sensors:** Detection of carbon monoxide, carbon dioxide, and hydrogen cyanide to monitor atmospheric conditions around the ladle and provide early warning of gas hazards.
- **Vibration Sensors:** Monitoring of mechanical vibrations in the ladle car structure to detect bearing wear, track irregularities, and structural integrity issues.
- **Weight/Load Cells:** Direct measurement of the mass of molten metal remaining in the ladle, providing a more accurate measure of fill level than indirect estimation methods.
- **Atmospheric Humidity Sensors:** Monitoring of moisture levels in the ladle vicinity, as moisture contact with molten metal can cause explosive steam generation.

9.6 INDUSTRY 4.0 INTEGRATION

The ultimate vision for SOLEIL is full integration into an Industry 4.0 framework, where the TLC monitoring data is seamlessly integrated with the plant's cyber-physical system infrastructure. This would enable closed-loop automated control of TLC operations, integration with production scheduling systems, real-time digital twin simulation of ladle operations, and coordinated multi-ladle optimization across the entire steel plant.

CHAPTER 10

CONCLUSION

This project has presented SOLEIL (Smart Observation & Ladle Efficiency with Intelligent Logic), a comprehensive embedded smart monitoring and control system designed to address the critical safety and efficiency deficiencies of manual Torpedo Ladle Car operations in steel manufacturing environments. The system was motivated by the significant risks—including over-tilting, spillage, overheating, and operator injury—associated with the absence of real-time, automated monitoring in existing TLC operations.

The SOLEIL system integrates the ESP32 microcontroller as the central processing and communication hub, the BNO055 9-axis gyroscope sensor for real-time tilt angle measurement, and MLX90614 infrared temperature sensors for non-contact thermal monitoring of the ladle. A threshold-based intelligent safety logic engine continuously evaluates sensor data against configurable safety limits and generates appropriate multi-modal alerts through LED indicators, audible buzzers, and an OLED display. The system's wireless communication capability, enabled by the ESP32's native Wi-Fi and Bluetooth radios, allows real-time remote monitoring and supervisory oversight without requiring operators to be physically present near the ladle.

Prototype validation demonstrated highly encouraging performance across all key metrics. Tilt angle measurement accuracy exceeded 98.5% across the full operational range, and temperature measurement accuracy consistently exceeded 97%. The safety logic achieved 100% correct classification with zero false positives or negatives in laboratory testing. Alert response times averaged 87 milliseconds, well within the sub-200 millisecond

target. Wireless communication demonstrated reliable performance with latencies below 70 milliseconds at 30-meter range.

The SOLEIL system addresses a genuine and pressing industrial safety need in a cost-effective, technically sound, and practically deployable manner. By transforming the Torpedo Ladle Car from a manually monitored, hazard-prone system into an intelligently monitored, data-driven, and automated safety system, SOLEIL contributes meaningfully to the broader goal of creating safer, more efficient, and more sustainable steel manufacturing environments.

The project also lays a strong technical foundation for future enhancements including AI/ML predictive analytics, advanced level sensing, cloud integration, mobile application development, and full Industry 4.0 integration. These future directions position SOLEIL as a platform technology with substantial long-term development potential, rather than a one-time prototype.

In conclusion, SOLEIL demonstrates that sophisticated, multi-parameter industrial safety monitoring can be achieved with commercially available, cost-effective embedded hardware, making advanced safety solutions accessible to a broader range of industrial operators than has previously been possible. The project serves as a compelling proof of concept for the application of smart embedded systems to industrial safety challenges in the steel sector and beyond.

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APPENDIX 1

SAMPLE ARDUINO FIRMWARE CODE – SENSOR INITIALIZATION

```
#include <Wire.h>
#include <Adafruit_BNO055.h>
#include <Adafruit_MLX90614.h>
#include <WiFi.h>

Adafruit_BNO055 bno = Adafruit_BNO055(55);
Adafruit_MLX90614 mlx = Adafruit_MLX90614();

const float TILT_WARNING = 45.0;
const float TILT_CRITICAL = 60.0;
const float TEMP_WARNING = 150.0;
const float TEMP_CRITICAL = 200.0;

void setup() {
  Serial.begin(115200);
  Wire.begin();
  if (!bno.begin()) {
    Serial.println('BNO055 init failed!');
    while (1);
  }
  if (!mlx.begin()) {
    Serial.println('MLX90614 init failed!');
    while (1);
  }
  Serial.println('SOLEIL System Initialized');
}

void loop() {
  imu::Vector<3> euler =
bno.getVector(Adafruit_BNO055::VECTOR_EULER);
  float tiltAngle = euler.x();
  float ambientTemp = mlx.readAmbientTempC();
  float objectTemp = mlx.readObjectTempC();
  checkSafetyThresholds(tiltAngle, objectTemp);
  delay(100); // 10 Hz sampling rate
}
```

APPENDIX 2

PIN CONFIGURATION TABLE – ESP32 TO PERIPHERAL MAPPING

ESP32 Pin	Component	Signal	Notes
GPIO 21 (SDA)	BNO055 / MLX90614 / OLED	I2C Data	Shared I2C bus with 4.7k Ω pull-up
GPIO 22 (SCL)	BNO055 / MLX90614 / OLED	I2C Clock	Shared I2C bus with 4.7k Ω pull-up
GPIO 2	Green LED	Status LED	Current limiting resistor 330 Ω
GPIO 4	Yellow LED	Warning LED	Current limiting resistor 330 Ω
GPIO 5	Red LED	Critical LED	Current limiting resistor 330 Ω
GPIO 18	Buzzer	PWM Alert	Via BC547 transistor driver
GPIO 19	Relay/Shutdown	Emergency Stop	Controls TLC power circuit
3.3V	All Sensors	Power	Regulated 3.3V rail
GND	All Components	Ground	Common ground

— End of Report —